

Benchmark choice and quotation persistence in county maize–urea price monitoring: Evidence from Liaoning, China

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Abstract

Benchmark choice changes agricultural price-monitoring signals, and persistent quotations affect both the historical threshold and later classifications. We analyze Liaoning maize–urea quotations over 2018–2026 using five benchmarks on the same 204 complete monitoring dates. Aggregate attainment for all twelve counties ranges from 34.31% to 99.02%, and aggregate alerts for the six screened counties range from zero to 131 dates. These differences persist under a common 50% county-share threshold and all twelve single-county deletions. The quotation screen flags six of twelve counties. Jianping records 109 identical urea quotations over 1,096 days; Jianping and Qingyuan account for 54.64% of accumulated shortfalls under the 2018–2020 benchmark. Four representative unchanged-urea runs overlap that historical window, linking quotation persistence to benchmark construction. All five paired excess-incidence intervals are positive for the original twelve counties. After excluding Jianping and Qingyuan, the 2020 interval is positive and the other four include zero; after the full screen, all five include zero. Historically flagged counties update less frequently in both later periods, while every long unchanged run coincides with updating same-product peers. The results show that benchmark selection and quotation history determine the monitoring signal. Reports should state the historical window, county membership and alert threshold alongside the quotation histories.

Keywords

Agricultural price monitoring; historical benchmarks; quotation persistence; maize–urea exchange ratio; county incidence; monitoring alerts

Highlights

- All twelve counties attain benchmarks on 34.31%–99.02% of complete dates.
- The combined quotation screen flags six of the twelve counties.
- Jianping and Qingyuan account for 54.64% of 2018–2020-benchmark shortfalls.
- For all twelve counties, 4/45 spells ≥ 3 observations have no updates (2018–2020).
- Historically flagged counties update less in both later observation windows.

1 Introduction

Agricultural price monitors turn reported quotations into signals that guide investigation of local price conditions. Two choices shape those signals: the historical prices used as a benchmark and the updating histories of the component quotations. Repeated values enter both the historical threshold and the later price index. We quantify how historical-window choice and county membership change the classifications and alert dates generated from the same price records.

An output–input price ratio relates the quoted value of a crop to an input cost. The World Bank's fertilizer-affordability measure compares fertilizer and food prices with a historical average (World Bank, 2024). We follow maize purchase and urea quotations jointly at county level and compare each county with its own historical exchange terms. Our benchmark defines a classification threshold, whereas an index reference period used solely to

rescale a series leaves relative movements unchanged (ILO et al., 2020). FAO's price-anomaly indicator instead compares quarterly and annual growth with month-specific historical distributions (Baquedano, 2015). We test historical-window choice and month-specific comparison separately to identify changes in the reference level and seasonal alignment.

The observation unit is a monitoring quotation. The meaning of a repeated value depends on its observation basis: a current local price, an earlier reported quotation, or a sampling point awaiting new price information. Measurement studies show that the observation process affects the apparent frequency of price changes (Eichenbaum et al., 2014). The Consumer Price Index Manual treats carrying a previous price forward as a procedure for handling missing information (ILO et al., 2020). We use product-specific updating, equal values across source changes and national-turn timing to identify persistent reporting patterns in the county archive.

China's grain-price monitoring rules specify observations on the fifth, fifteenth and twenty-fifth days of each month and assign review and aggregation responsibilities (National Development and Reform Commission, 2011). Agricultural-input rules distinguish sampling-day transactions, posted prices for available stock and the latest transaction when an item is temporarily unavailable (Price Monitoring Center, 2015). Liaoning's monitoring-point review introduced trial reporting, site visits and replacements in response to collection difficulties (Liaoning Provincial Development and Reform Commission, 2022). These procedures make the sampling enterprise and the date of its underlying price information relevant to repeated quotations.

The reference windows cover three complete archive years after the 2016 maize procurement reform and before the 2021 fertilizer-price surge. We use the full 2018–2020 window, remove each endpoint year in turn, compare with the closest complete year, and match observations by calendar month. This design separates changes in the historical level from within-year variation. For agricultural information services, the usefulness of information and the design of delivery are central concerns (Shepherd, 1997; Galtier et al., 2014). Alert dates and episodes translate the benchmark differences into the signals a monitoring service would review.

We show that benchmark selection changes attainment and alert frequency even after every single-county deletion. The quotation screen identifies persistent county–product records that enter the historical reference as well as the later index. Historically flagged counties continue to update less frequently in later periods, and same-product quotations change elsewhere in the province during every long unchanged run. These findings connect benchmark construction to the reporting histories behind the monitoring signal.

2 Institutional background and economic framework

China replaced temporary maize procurement in the northeastern provinces and Inner Mongolia with market-oriented purchasing and subsidies in 2016 (National Development and Reform Commission, 2017). The 2018–2020 window supplies three complete archive years after that reform and before the 2021 fertilizer-price surge. In 2021, domestic fertilizer-stabilization measures addressed production inputs, reserves, transport and supervision, while mandatory export inspection for 29 ten-digit fertilizer customs codes took effect on 15 October (National Development and Reform Commission and other departments, 2021; General Administration of Customs, 2021). These dated changes define the historical windows and later price environment relevant to the study.

Fertilizer shocks provide the economic setting. Hu et al. (2025) separate energy-related and urea-specific components of Chinese price movements; Willwerth et al. (2025) find heterogeneous Kenyan agro-dealer responses to import-price increases and decreases. We track the input quotation together with maize to determine which recorded component changes accompany improvement in local exchange terms.

Let $P_{C,it}$ and $P_{U,it}$ denote the maize purchase and domestic-urea quotations in the same currency per kilogram. Their exchange ratio is

$$Q_{it} = \frac{P_{C,it}}{P_{U,it}}. \quad (1)$$

The ratio represents kilograms of urea at the quoted monetary value of one kilogram of maize. Fertilizer expense as a share of maize revenue equals $(u/Y)/Q$, where u is fertilizer quantity and Y is maize output sold. Fertilizer profitability combines yield response with output and input prices (Ragasa et al., 2025); Northeast Chinese spring-maize recommendations use soil-fertility groups and response functions (Bao et al., 2020). The exchange ratio measures the quoted relative-price component of this relationship.

An unchanged urea quotation gives $\Delta \log Q_{it} = \Delta \log P_{C,it}$: maize supplies the recorded movement in the exchange ratio, and the repeated urea quotation sets its level. Repeated urea values within the historical window also enter the benchmark. The product histories therefore link changes in the exchange ratio to both current price adjustments and the historical comparison level.

3 Data and measurement

3.1 Archive, calendars and county groups

The Liaoning archive records mixed-grade maize purchase and domestic-urea quotations from 5 January 2018 to 25 September 2026 (Liaoning Provincial Development and Reform Commission, n.d.). The normalized data contain 9,450 records over 315 monitoring dates: 8,987 positive values, 453 original zeros and ten missing values. Fifteen geographic slots appear per date, with sixteen identifiers across the full record. Product definitions, county identifiers, original numerical strings, observation states and source labels are recorded. We calculate logarithms from positive price pairs.

The original twelve counties are those shared by the historical comparison endpoints. Their 2018–2020 calendar has 108 complete dates. The later calendar has 207 scheduled observations, of which 204 are complete. We also report eleven counties excluding Jianping as a sensitivity group. Removing Jianping and Qingyuan through the low-update branch leaves ten counties. The combined screen also removes Sujiatun, Dashiqiao, Zhangwu and Dawa, leaving Faku, Zhuanghe, Fengcheng, Dengta, Changtu and Kaiyuan. This six-county group provides the comparison after the complete screen. We recalculate each group’s aggregate, attainment dates and county residuals on the original completeness mask, so membership comparisons use the same observation calendar.

Individual shortfall sums use all valid county pairs, including observed county cells when the aggregate is unavailable. Update-rate denominators require a positive scheduled predecessor and source continuity. Constant-price histories use each series’ positive observations, with gaps interrupting a sequence.

Table 1. Data coverage, historical benchmarks and county groups.

Component	Group or window	Coverage and meaning
County archive	Fifteen geographic slots; sixteen historical IDs	315 dates; 9,450 records; original zero and missing states retained
Historical calendar	January 2018–December 2020	108 complete monitoring dates

Component	Group or window	Coverage and meaning
Later aggregate comparisons	January 2021–September 2026	207 scheduled dates; common 204-complete-date mask
B0	2018–2020	County mean log exchange ratio; 108 historical dates
B1	2019–2020	County mean log exchange ratio; 72 historical dates
B2	2020	County mean log exchange ratio; 36 historical dates
B3	2018–2019	County mean log exchange ratio; 72 historical dates
B4	Month-specific 2018–2020	Twelve monthly county means, based on 108 dates
Full12	Original twelve counties	Original monitoring cohort
Eleven counties	Exclude Jianping	Sensitivity and selected precision comparisons
U10	Exclude Jianping and Qingyuan	Group defined by the low-update branch
S6	Faku, Zhuanghe, Fengcheng, Dengta, Changtu, Kaiyuan	Counties passing the combined low-update or run-amplitude screen
Individual B0 shortfalls	Full12 valid county pairs	2,481 valid county-period pairs, including dates with an incomplete aggregate

Note: Monitoring periods refer to scheduled archive observations. Aggregate comparisons use the same 204 complete dates within the 207-slot later calendar. County benchmark means use the same historical dates for every group; changing membership recalculates the aggregate, attainment set and county residuals. The month-specific benchmark uses twelve monthly means per county.

3.2 Original values and external comparison

County quotations are expressed in CNY per 500 g. Multiplication by two converts them to CNY per kilogram and preserves their ratio. The domestic-urea field identifies the product and unit; its transaction-stage designation is unspecified. We identify equal quotations and updates from their original decimal values. Screening runs stop at the source change between 25 January and 5 February 2024. Descriptive numerical-persistence segments continue across that boundary when consecutive scheduled positive values remain equal. Zeros and missing planned observations end a continuous observed segment; equal values separated by a gap receive a separate observed-span record.

National Bureau of Statistics circulation-survey reports supply second-grade yellow-maize and small-/medium-small-particle urea quotations in CNY per tonne (National Bureau of Statistics of China, 2024a, 2024b). Their wholesale scope and specifications differ from the county fields. We compare proportional changes within each series, which places their movements on the same dimensionless scale. Endpoints are matched to nominal dekad dates, the fifth, fifteenth and twenty-fifth; unavailable national endpoints remain missing. Matching national endpoints to the middle of each reporting interval leaves all screening flags unchanged.

We use proportional national price movement to compare unchanged county quotations with the wider market. An observed national amplitude above 20% records substantial movement in the matched national series while the county quotation stays fixed. The maximum-to-minimum range across observed national prices is a lower bound on the full range over the run when national coverage is partial.

The source archive contains 610 responses, including six original HTTP failures. Nineteen selected original-page checks link normalized records back to the reported cells. Additional commodities are grouped by source-local identity, exact specification and unit, with capture coverage recorded for each group. The complete observation records and date-level results accompany the analysis as replication materials.

3.3 Benchmarks and measurement

The 2018–2020 benchmark uses each county's mean log exchange ratio over all three complete reference years. It averages historical exchange terms across three annual cycles after the maize procurement reform and before the 2021 fertilizer-price surge.

The two-year windows remove the endpoints symmetrically. The 2019–2020 benchmark excludes 2018 and tests the influence of the earliest reference year; the 2018–2019 benchmark excludes 2020 and tests the influence of the latest. The 2020 benchmark supplies the closest complete annual comparison before the 2021 fertilizer shock. This timing rationale matters because Liaoning's autumn-grain survey reports a maize purchase price of 1.18 CNY per 500 g on 25 November 2020, compared with 0.90 a year earlier (Liaoning Provincial Development and Reform Commission, 2020).

The month-specific 2018–2020 benchmark compares each later observation with the same calendar month in the three reference years. We quantify the contribution of month indicators to separate within-year variation from differences in historical levels. Table 1 gives abbreviated benchmark and group labels for the tabulated results.

For county benchmark b_i , define

$$\begin{aligned} g_{it} &= \log Q_{it} - b_i, \quad I_{it} = \exp(g_{it}), \\ G_t &= \frac{1}{N} \sum_{i=1}^N g_{it}, \quad A_t = \exp(G_t), \\ H_t &= \frac{1}{N} \sum_{i=1}^N 1\{g_{it} < 0\}. \end{aligned} \tag{2}$$

For the month-specific benchmark, b_i takes the target month's value. Aggregate attainment means $G_t \geq 0$. Mean H_t on attainment dates measures the share of counties below their own benchmarks while their aggregate attains its benchmark. County shortfalls sum $\max(-g_{it}, 0)$ across valid observations and have units of log-point monitoring periods. We classify aggregate attainment date by date; the alert rule in Section 4.5 starts after three consecutive qualifying observations.

4 Empirical strategy

4.1 Historical-residual reference along the observed aggregate path

We hold the observed later aggregate path constant and replace county deviations with historical residual vectors. On donor date s ,

$$r_{is} = g_{is} - G_s, \quad g_{it}^{ref} = G_t + r_{i,s(t)}. \tag{3}$$

We sample whole county-date vectors in non-circular moving blocks, keeping cross-county co-movement and short-run dependence together (Künsch, 1989). Six monitoring periods form the main block length. Each configuration uses 1,999 draws on the scheduled calendar, followed by the common completeness mask. Observed

and reference statistics use the same attainment dates. We compare observed below-benchmark incidence with the distribution generated by historical county deviations along the observed aggregate path.

We report the mean reference incidence and its two-sided 90% interval, from the fifth to the ninety-fifth percentile. The one-sided upper-tail estimate is $\hat{p} = (K + 1) / (R + 1)$, where K counts reference statistics at least as large as the observation and R counts valid draws. Ties are included. The fixed reference varies county deviations at the estimated historical threshold; the paired comparison below also varies the historical benchmark. Appendix A.1 gives the calendar alignment used in the annual and month-specific comparisons.

4.2 Paired benchmark re-estimation

We resample historical county-date vectors and re-estimate each county's benchmark. Each draw recalculates the later aggregate path and attainment dates. Historical residuals are recomputed relative to the re-estimated benchmarks and centered across counties on every donor date. For each benchmark draw, we sample one independent historical residual path and compare both incidence statistics on the recalculated attainment dates.

Let O^* and K_{ref}^* denote the observed and reference below-benchmark county-date counts, and T^* the attainment-date count. The paired difference is

$$D^* = \overline{H}^{obs,*} - \overline{H}^{ref,*} = \frac{O^* - K_{ref}^*}{N T^*}. \quad (4)$$

We report the mean difference, a two-sided 95% percentile interval and the share of draws with $D^* > 0$. Each paired draw combines re-estimated historical means with a sampled historical residual path. The mean and interval therefore summarize both sources of variation. We identify excess incidence with a paired interval above zero and classify intervals containing zero separately.

The fixed reference isolates county-level residual variation; the paired comparison adds benchmark estimation. For the ten-county group under the 2019–2020 benchmark, the fixed-reference tail estimate is 0.010. Benchmark re-estimation produces a paired interval containing zero, showing that historical-mean variation changes the classification of excess incidence. Table 3 compares the three main county groups on identical dates. Appendix A.1 gives the resampling conventions; complete block-length results accompany the replication materials.

4.3 Resampling precision

We use 1,999 draws for every comparison in Table 3 and extend six selected comparisons to 9,999 draws: the 2018–2020 benchmark for twelve, eleven and six counties; the 2020 benchmark for ten and six counties; and the month-specific benchmark for eleven counties. Section 5.2 reports the extended comparisons. Appendix A.2 defines tail estimates, Monte Carlo standard errors and percentile conventions.

4.4 Quotation screening and episode updates

The low-update branch flags counties with either product's update rate strictly below one-third of that product's median across the original twelve counties. Rates divide changes in recorded decimal values by positive, scheduled, same-source adjacent comparisons. The median remains tied to the original twelve-county group as counties are removed. The second branch flags source-specific constant runs longer than 36 observations with matched national price amplitude strictly above 20%. Amplitude is $\max(P) / \min(P) - 1$, evaluated from the original decimal prices.

An observed national range exceeding the amplitude threshold flags the run-amplitude branch even with partial coverage. Lower observed ranges with unobserved periods receive an unresolved classification. Either screening branch flags a county, so the low-update branch classifies counties independently of national coverage. The threshold grid combines update-rate fractions of one-quarter, one-third, one-half and two-thirds; minimum run lengths strictly above 24, 36 and 48 observations; and national-amplitude thresholds strictly above 10%, 20% and 30%. These choices give 36 settings. We apply each setting to all twelve counties and recalculate the comparisons for each resulting county group.

Below-benchmark spells under the 2018–2020 benchmark end at a changed classification, invalid price pair or source boundary. A spell of length ℓ has $\ell - 1$ internal comparison opportunities. Updates compare quotations within the spell and exclude the entering transition. We group spells by length and calculate internal update rates and zero-update shares for spells containing at least two observations. County rates pool changes and eligible adjacent comparisons.

4.5 Monitoring consequences

Rule i starts on the third consecutive complete scheduled date with $G_t < 0$ and covers subsequent consecutive below-benchmark dates. Rule ii triggers on $3k_t \geq N$, where k_t counts below-benchmark counties. Its minimum counts are four for the original twelve-county group, four for the ten-county group and two for the six counties passing the combined screen. A common 50% sensitivity uses $2k_t \geq N$, requiring six, five and three counties, respectively.

Missing scheduled observations reset the aggregate streak. A non-triggering or missing date ends an episode; consecutive triggering dates form an episode across source boundaries. Trigger fractions use 204 complete dates, and the date-level records include all 207 slots.

4.6 Monthly variation and quotation timing

We measure monthly variation in the complete 2018–2020 panel, which has 108 historical dates and 1,296 county–date pairs for all twelve counties. The six screened counties supply 648 pairs. The primary outcome is $\log Q$, with equal weight for each valid county–date pair. We estimate a month-indicator model, a county-and-year model, and a county-year-month model. Let $SS E_M$, $SS E_R$ and $SS E_F$ denote their residual sums of squares, respectively, and let TSS be the sum of squares centered on the overall outcome mean. The raw month share is $1 - SS E_M / TSS$; the incremental share is $(SS E_R - SS E_F) / TSS$; the partial share divides the same reduction by $SS E_R$. Level ratios, equal-date weights and the six-county panel provide measurement checks.

For quotation timing, we average observed national urea prices by month without interpolation. The primary rule requires at least two observed dekads per month, an opposite preceding movement, two consecutive movements in the new direction, and a cumulative change of at least 5% from the extremum. Flat months, insufficient monthly coverage and calendar gaps interrupt the direction sequence. A national turn becomes available on the latest publication date of the observations needed to confirm it. Each county update is matched to the most recent same-direction turn available by its monitoring date. Three- and six-month distances are measured from the extremum month. A 10% threshold and complete three-dekad coverage provide checks. The rules apply to all twelve counties; Appendix A.3 lists all six Jianping urea changes.

4.7 County influence and comparisons across time and place

We delete each of the twelve counties once under all five benchmarks, giving sixty comparisons in addition to the original twelve-county controls. Every deletion uses the original 204 complete dates, six-period blocks and the same 1,999 sampling paths. County means, residuals and aggregate classifications are recalculated for the eleven remaining counties. This design reports the influence of every county under every benchmark.

We then apply the quotation screen to county observations from 5 January 2018 to 25 December 2020, matching national information published by that endpoint. The update-rate cutoff is one-third of each product's historical twelve-county median; the run branch uses more than 36 observations and national amplitude above 20%. County membership is fixed before comparing 2021–2023 and January 2024–September 2026. Update rates pool changes and eligible same-source adjacent transitions within each period. Later alerts use the 2018–2020 county benchmarks and the original complete-date mask.

Finally, every same-source unchanged run longer than 36 observations is compared with the same product, specification and unit in the other eleven original counties. The target county is excluded from this fixed comparison group. We calculate its median on dates with positive quotations from all eleven peers. We report median amplitude, complete-date coverage and the number of peers with at least one eligible update during each run.

5 Results

5.1 The complete screen flags six counties

The low-update branch identifies Jianping for maize and urea, and Qingyuan for urea. Jianping changes its maize quotation in 35 of 311 valid comparisons and its urea quotation in six of 311; Qingyuan changes maize in 48 of 313 (15.34%) and urea in eight of 313 (2.56%). Both Jianping products fall below their product-specific low-update cutoffs, whereas Qingyuan enters the branch through urea. The county-series medians are 37.40% for maize and 11.18% for urea, giving one-third thresholds of 12.47% and 3.73%. These product-specific rates define membership of the ten-county group.

Adding the run-amplitude branch also flags Sujiatun, Dashiqiao, Zhangwu and Dawa. The observed national maximum-to-minimum amplitude is 20.87%–42.41% across the six representative unchanged-quotation windows, exceeding the 20% cutoff in every case (Table 2). The complete screen flags six counties and leaves the six-county sample named in Section 3.1.

Table 2. Product and run evidence for the six counties flagged by the complete screen.

Flagged county	Low-rate evidence	Run-amplitude product	Representative source-specific run	Observations	Observed national amplitude
Sujiatun	None	Urea	2018-01-05–2020-04-15	83	29.03%
Qingyuan	Urea: 8/313 (2.56%)	Urea	2018-01-05–2020-03-15	80	29.03%
Dashiqiao	None	Urea	2019-05-05–2021-02-25	66	33.35%
Zhangwu	None	Urea	2025-02-25–2026-09-25	58	20.87%
Dawa	None	Urea	2024-02-05–2025-03-15	41	42.41%
Jianping	Maize: 35/311 (11.25%); urea: 6/311 (1.93%)	Urea	2018-04-25–2021-04-25	109	33.53%

Note: The low-rate branch uses county-series medians across all twelve counties. Every representative run is source-specific and longer than 36 observations. National amplitude is the observed max/min–1; all six representative windows have partial coverage and an observed lower bound above 20%. Jianping's low-rate products are maize and urea; its representative constant run is urea. The replication data contain all eleven triggering runs and their matched national windows.

Zhangwu’s observed national amplitude of 20.87% exceeds the 20% cutoff by 0.87 percentage points, making its flag sensitive to the amplitude threshold (Table 2). Its 58-observation urea run ends on 25 September 2026, the archive endpoint, and is right-censored. Across the threshold grid, Zhangwu is confirmed as flagged in all 24 settings using 10% or 20% amplitude thresholds and is unresolved in the twelve settings using 30%, across every update-rate fraction and run-length cutoff examined (the complete threshold grid in the replication materials).

Four of the six representative urea runs fall within or span the 2018–2020 benchmark window: Sujiatun and Qingyuan lie within it, while Dashiqiao and Jianping extend into 2021 (Table 2). Persistent quotations thus enter historical benchmark construction as well as subsequent index histories.

5.2 Benchmark and county membership change the distributional evidence

Benchmark choice shifts aggregate attainment for all twelve counties between 34.31% (70 of 204 dates) under 2020 and 99.02% (202 of 204 dates) under 2018–2019 (Table 3; Figure 1). Among aggregate-attainment dates, mean county incidence H ranges from 11.18% to 24.40%. The result under the three-year benchmark is 17.05%, compared with a fixed-reference mean of 13.38%. Its paired interval is [0.20, 5.12] percentage points. The original twelve-county group’s paired interval is strictly positive under each of the five benchmarks.

Table 3. Five benchmarks and three county groups: six-period blocks and 1,999 draws.

Bench mark	Group	Attainment dates	Observed H (%)	Reference H (%)	Fixed p (MCSE)	Mean D (pp)	Paired 95% interval (pp)	D>0 (%)
B0	Full12	174	17.05	13.38	$\leq 0.0005 \dagger$	2.66	[0.20, 5.12]	98.50
B0	U10	185	11.78	11.97	0.578 (0.011)	-0.74	[-2.91, 1.48]	23.91
B0	S6	179	10.06	12.15	0.971 (0.004)	-2.63	[-5.62, 0.00]	2.20
B1	Full12	153	19.93	16.52	$\leq 0.0005 \dagger$	4.61	[1.41, 8.77]	99.80
B1	U10	159	15.41	13.52	0.010 (0.002)	2.55	[-0.97, 8.36]	88.64
B1	S6	146	12.33	13.64	0.906 (0.007)	-0.80	[-4.04, 3.37]	28.66
B2	Full12	70	24.40	17.28	$\leq 0.0005 \dagger$	8.58	[3.10, 14.06]	100.00
B2	U10	79	24.56	15.05	$\leq 0.0005 \dagger$	9.42	[2.89, 14.00]	99.45
B2	S6	66	18.69	17.39	0.235 (0.009)	2.68	[-3.96, 8.16]	83.69
B3	Full12	202	11.18	6.44	$\leq 0.0005 \dagger$	4.79	[2.61, 7.15]	100.00
B3	U10	202	3.86	5.04	0.975 (0.004)	-0.49	[-2.65, 1.72]	33.17
B3	S6	203	3.12	6.48	$\geq 0.9995 \dagger$	-2.56	[-5.83, 0.57]	5.50
B4	Full12	172	16.91	12.66	$\leq 0.0005 \dagger$	4.40	[2.08, 6.89]	100.00
B4	U10	178	10.96	10.20	0.110 (0.007)	1.63	[-0.70, 4.54]	89.94
B4	S6	181	11.97	11.71	0.390 (0.011)	0.58	[-2.27, 3.48]	64.38

Note: H is a percent; D and its interval are percentage points. Fixed-reference intervals are two-sided 90%; paired intervals are two-sided 95%. Fixed-reference p is a one-sided upper-tail estimate with ties, with MCSE in parentheses. Boundary entries are $\leq 0.0005 \dagger$ for a zero tail count and $\geq 0.9995 \dagger$ for a complete tail count; Appendix A.2 gives their binomial bounds. D>0 is the share of positive paired draws. Paired intervals combine variation in historical means and in the residual path sampled for each draw. Table 1 defines the benchmark and county-group labels.

Excluding Jianping and Qingyuan changes the excess-incidence pattern. The ten-county group has a positive paired interval under the 2020 benchmark, [2.89, 14.00] percentage points. The 2019–2020 fixed-reference estimate is 0.010, and the paired interval is $[-0.97, 8.36]$. Re-estimating historical means therefore changes the classification of excess incidence. The three-year, 2018–2019 and month-specific intervals also include zero.

The six counties passing the combined screen have paired intervals including zero under all five benchmarks. Their mean difference ranges from -2.63 percentage points under the 2018–2020 benchmark to 2.68 under the 2020 benchmark. For 2020, the fixed-reference estimate is 0.235 and the paired interval is $[-3.96, 8.16]$. For 2018–2020, the 1,999-draw upper endpoint is exactly zero. The full screen changes the excess-incidence pattern across the historical benchmarks: the original twelve-county intervals are positive, and the screened six-county intervals all include zero.

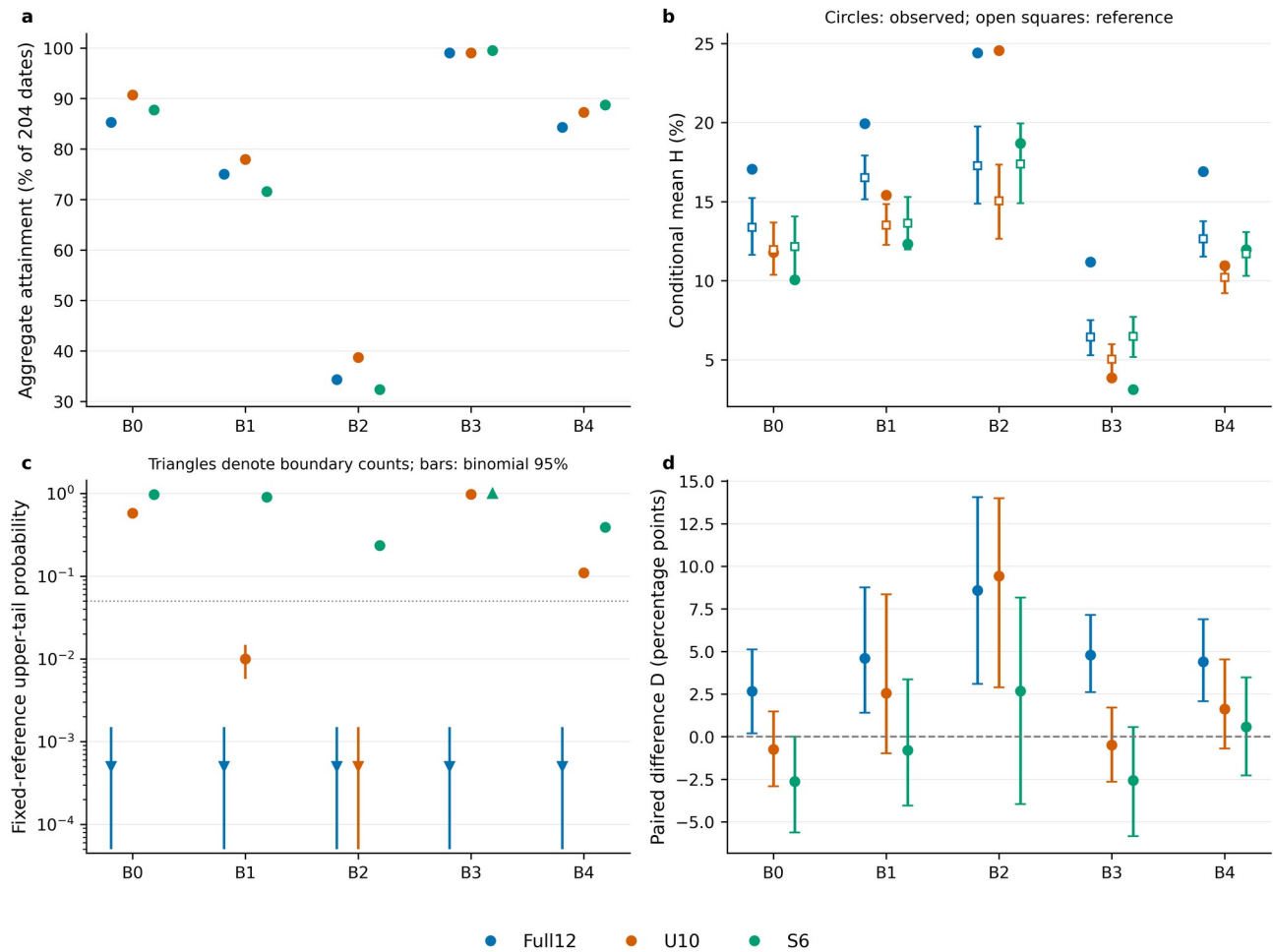


Figure 1. Benchmark dependence for all twelve counties, ten counties excluding Jianping and Qingyuan, and the six screened counties. Panels show aggregate attainment on the same 204 complete dates, mean county incidence H on aggregate-attainment dates and the fixed-reference 90% interval, the fixed-reference tail estimate with Monte Carlo uncertainty, and paired mean D with its 95% interval. Benchmarks are categorical; points are grouped by county membership. All panels use six-period blocks and 1,999 draws. H is a percent; D is in percentage points. Source: the five-benchmark comparisons summarized in Table 3.

The selected 9,999-draw checks give intervals of $[0.18, 5.12]$ percentage points for all twelve counties under the 2018–2020 benchmark and $[3.04, 14.03]$ for the ten-county group under the 2020 benchmark. The six-county intervals are $[-5.72, 0.09]$ and $[-3.71, 8.07]$, respectively. In the former, the upper endpoint moves from exactly zero at 1,999 draws to above zero at 9,999 draws. The complete precision results include all six selected comparisons, including the eleven-county sensitivity cases.

5.3 Persistent quotation levels determine which product moves the ratio

Under the 2018–2020 benchmark, accumulated shortfall is 47.54 log-point monitoring periods. Jianping contributes 22.71 and Qingyuan 3.27, together accounting for 54.64% of the unrounded total. The other ten counties contribute 21.56. These sums use valid county observations, including dates without a complete aggregate.

Jianping reports 1.00 CNY per 500 g for urea on 109 consecutive observations from 25 April 2018 to 25 April 2021, spanning 1,096 elapsed calendar days. The 2018–2020 benchmark also includes eleven earlier quotations of 0.75, so it combines two recorded urea levels. Urea rises to 1.10 on 5 May 2021 and to 1.40 on 15 August 2021. The domestic specification and quotation unit remain the same across the latter change.

A later urea level of 1.50 persists for 94 observations from 15 April 2022 to 15 November 2024, crossing the archive source boundary. A missing scheduled observation on 25 November separates this sequence from another 44 equal observations between 5 December 2024 and 15 February 2026. The 138 equal quotations form two consecutive segments of 94 and 44 observations, separated by the missing November 2024 slot. During the first segment, the national urea quotation falls from 2,885.2 to 1,867.2 CNY per tonne, a decline of 35.28%; the other-eleven-county endpoint median falls from 1.60 to 1.00 CNY per 500 g.

Qingyuan's longest urea segment contains 80 observations at 1.10 from 5 January 2018 to 15 March 2020. Its later 1.60 segment contains 75 observations from 25 March 2022 to 15 April 2024 and crosses the source boundary without a county gap. The corresponding other-eleven-county medians are 1.55 and 1.25 at its endpoints. Qingyuan's comparison group includes Jianping, while Jianping's group includes Qingyuan; their comparison lines have different missing-date masks. Figure 2 follows each observed series on its own calendar.

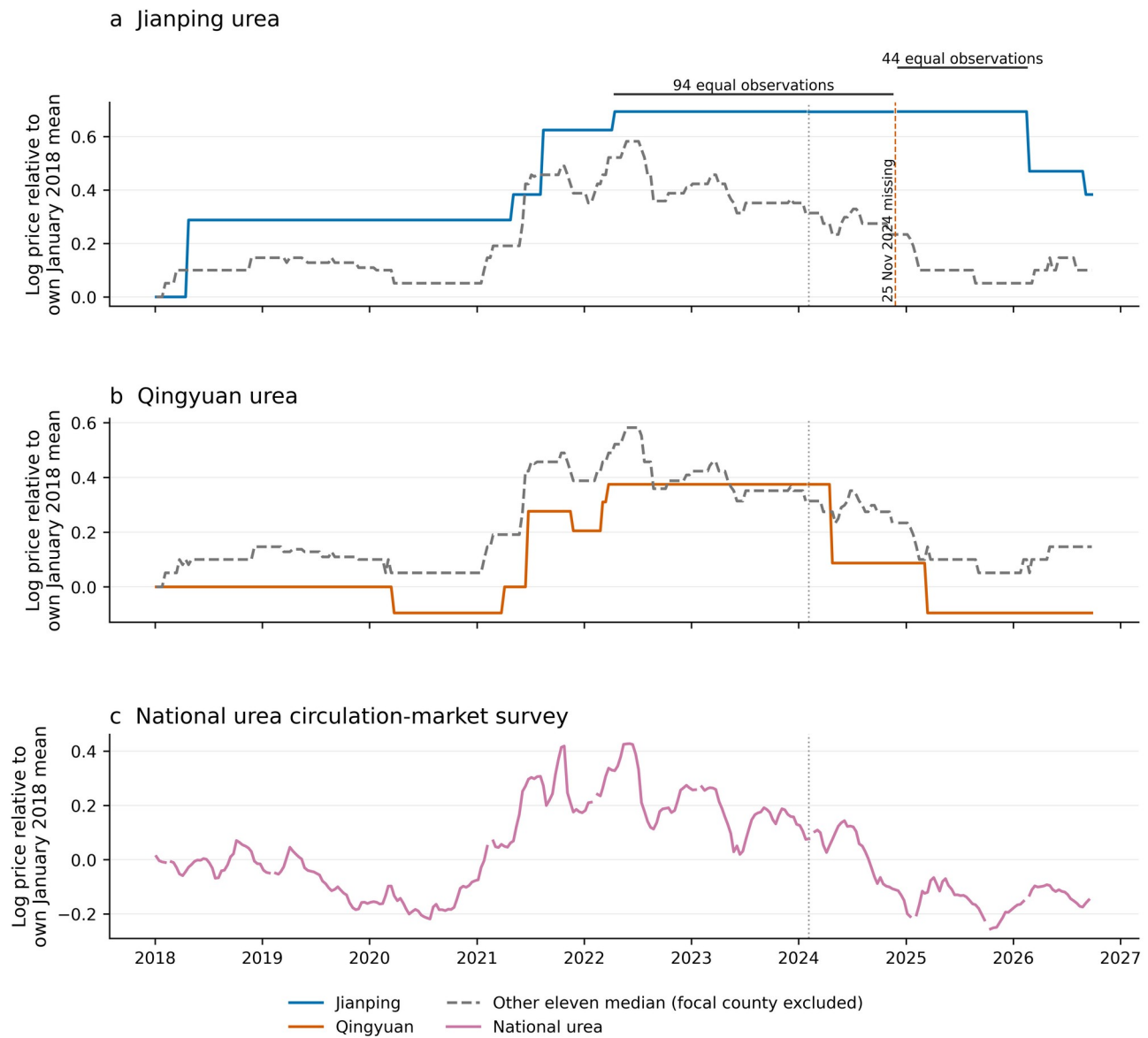


Figure 2. Urea paths for Jianping and Qingyuan, each compared with the other-eleven-county median, and national urea. Log prices are centered on each series' January 2018 mean. Lines break at source changes or unavailable observations. Each peer series follows its own complete-date mask. Sources: the archived county quotations and national circulation-survey observations.

The timing check includes all six source-continuous Jianping urea updates. Five have an earlier, already confirmed same-direction national turn under the primary 5% rule. Their distances from the extremum are eight, five, four, nine and five calendar months; the April 2018 update has no eligible preceding turn in the observed archive. Three of six updates fall within six months, and none falls within three. Raising the amplitude threshold to 10% gives two six-month matches; requiring all three national dekads per month gives one at 5% and zero at 10%. The confirmation-publication dates and unmatched update are reported alongside every local change (Appendix A). Jianping's urea quotations update infrequently and at dispersed dates while national prices change.

Recorded shortfalls generally include quotation changes. The twelve-county sample has 74 spells under the 2018–2020 benchmark, including nineteen single-observation spells with no internal comparison. Among spells lasting at least three observations, four of 45, or 8.89%, contain no update of either product. The complete episode records give every length group and opportunity-weighted rate.

The wider archive contains 31 exact product/specification/unit combinations. At the main screening setting, 19 of 372 county–product classifications are confirmed flags, 78 are confirmed non-flags and 275 are unresolved. The archive inventory records capture coverage, national-reference availability and threshold-grid classifications for every product group.

5.4 Benchmark differences persist under a common share threshold

Table 4 shows the alerts generated by different benchmarks and county samples. Under the 2018–2020 benchmark, aggregate alerts occur on nineteen dates for all twelve counties, twelve for the ten-county group and sixteen for the six counties passing the combined screen. Under the 2020 benchmark, the counts are 126, 116 and 131. The 2018–2019 benchmark gives no aggregate alerts in any group. The six-county sample therefore ranges from zero to 131 aggregate-alert monitoring dates on the same 204 complete dates.

Table 4. Trigger monitoring dates and episodes under three rules.

Benchmark	Group	Aggregate dates/episodes	H≥1/3 dates/episodes	H≥50% dates/episodes
B0	Full12	19/4	63/11	31/6
B0	U10	12/2	42/7	26/6
B0	S6	16/3	57/10	34/7
B1	Full12	38/5	102/5	62/11
B1	U10	33/5	77/9	56/10
B1	S6	38/7	95/6	68/10
B2	Full12	126/4	157/8	137/3
B2	U10	116/4	146/6	132/7
B2	S6	131/3	159/7	145/7
B3	Full12	0/0	15/3	7/3
B3	U10	0/0	11/4	5/3
B3	S6	0/0	13/4	5/2
B4	Full12	22/4	66/11	34/10
B4	U10	14/5	46/9	31/10
B4	S6	12/4	67/9	35/12

Note: Each cell reports monitoring dates/episodes. Aggregate alerts start on the third consecutive complete scheduled date below the benchmark. For the twelve-, ten- and six-county groups, respectively, the one-third rule requires 4/4/2 below-benchmark counties and the 50% rule requires 6/5/3. All percentages and classifications use the common 204 complete dates within the 207 scheduled slots. The replication records give all trigger dates, percentages and first/last triggers; Table 1 defines the table labels.

The common 50% threshold gives the six-county sample 34, 68, 145, five and 35 trigger dates under the 2018–2020, 2019–2020, 2020, 2018–2019 and month-specific benchmarks. The corresponding one-third counts are 57, 95, 159, thirteen and 67. The common county-share threshold leaves substantial differences in alert frequency across benchmarks. Every 50% trigger date also meets the one-third rule; additional episodes arise when the stricter threshold interrupts a run.

For all twelve counties under the 2019–2020 benchmark, the stricter share threshold reduces alert dates from 102 to 62 but raises episodes from five to eleven. For the six-county sample under the month-specific benchmark, dates fall from 67 to 35 and episodes rise from nine to twelve. Screening changes alerts in both directions: removing two low-update counties cuts 2018–2020 aggregate alerts from nineteen to twelve dates, while the full screen raises 2020 alerts from 126 to 131.

5.5 Historical levels and month matching

The county panel records different exchange terms across the reference years. For all twelve counties, annual geometric mean exchange ratios are 0.797 in 2018, 0.800 in 2019 and 0.997 in 2020. The 2020 mean is 24.60% above 2019, calculated from unrounded values. All twelve county-specific means rise in that comparison. The one-year 2020 benchmark therefore sets a higher comparison level than the earlier reference years.

Month indicators explain 4.97% of total pre-2021 log-ratio variation. After county and year effects, they explain 10.48% of the remaining variation; the county-and-year model explains 52.58% of total variation, and the full model explains 57.55%. The balanced panel makes the raw and incremental month shares equal. Equal-date weights give the same results. For the six screened counties, the raw and partial shares are 5.95% and 13.47%, respectively.

Month matching changes which monitoring dates qualify even when its net effect is small. For all twelve counties on the common 204-date calendar, the 2018–2020 benchmark has 174 attainment dates and the month-specific benchmark has 172. Eighteen dates change classification: ten leave the attainment set and eight enter it. Matching months thus changes the timing of the monitoring message as well as the total number of qualifying dates.

6 County influence and additional quotation diagnostics

6.1 Benchmark differences survive symmetric county deletion

The sixty deletion comparisons produce 53 strictly positive paired intervals and seven intervals containing zero (Table 5). Under 2018–2020, deleting Qingyuan, Dawa, Changtu or Jianping makes the interval include zero. Under 2019–2020, 2018–2019 and the month-specific benchmark, Jianping’s deletion produces an interval containing zero; the other eleven deletions produce positive intervals. Every deletion remains strictly positive under 2020. The sixty comparisons identify each county’s influence across all five benchmarks.

Table 5. Results across all twelve single-county deletions under each benchmark.

Historical benchmark	Strictly positive intervals	Intervals containing zero	Attainment range (%)	Aggregate alert dates
2018–2020	8	4	82.84–89.71	12–25
2019–2020	11	1	68.63–79.41	32–47
2020	12	0	31.37–37.25	118–134
2018–2019	11	1	97.06–100.00	0–2
Same month, 2018–2020	11	1	82.84–88.24	15–25

Note: Each row summarizes twelve single-county deletions on the same 204 complete monitoring dates. All comparisons use six-period blocks and 1,999 paths. Intervals are classified using unrounded endpoints. Aggregate alerts begin with the third consecutive below-benchmark scheduled observation; missing observations reset the sequence.

Benchmark choice continues to determine alert intensity after every county deletion. The 2020 benchmark produces 118–134 aggregate-alert dates, compared with zero to two under 2018–2019. Under the month-specific benchmark, the Jianping-deletion interval is [0.00, 5.01] percentage points, with its lower endpoint exactly at zero.

6.2 Historical quotation patterns continue in later periods

Information available through December 2020 flags Sujiatun, Qingyuan, Dashiqiao, Dawa and Jianping. Faku is unresolved because national coverage in its relevant run is incomplete; the other six counties have no historical flag. The later comparison fixes these five flagged counties and the seven without a confirmed flag, including Faku.

The historically flagged group has lower pooled updating for both products in both later windows (Table 6). Urea updating is 13.46% versus 25.23% in 2021–2023 and 6.42% versus 18.11% in January 2024–September 2026. Maize updating is 27.85% versus 56.57% and 23.19% versus 51.40%, respectively. The lower updating rates persist in both later periods for counties identified from the historical observations.

Table 6. Later update counts and pooled rates by fixed historical membership.

Period	Product	Five historically flagged counties	Seven without a confirmed historical flag
2021–2023	Maize	149/535 (27.85%)	422/746 (56.57%)
2021–2023	Urea	72/535 (13.46%)	189/749 (25.23%)
January 2024–September 2026	Maize	112/483 (23.19%)	349/679 (51.40%)
January 2024–September 2026	Urea	31/483 (6.42%)	123/679 (18.11%)

Note: Historical membership is fixed before the two later periods. The comparison group includes six counties without a historical flag and Faku, whose status is unresolved. Each cell gives changes divided by eligible same-source adjacent transitions. Percentages pool counts across counties; product missingness determines the denominators.

The same historical screen also changes later alert classifications. Under the 2018–2020 benchmark, all twelve counties have thirteen aggregate-alert dates in 2021–2023, compared with twelve for the seven-county comparison group. In January 2024–September 2026 the counts are six and four. Both groups use the original common calendar, with 106 and 98 complete dates in the two periods.

6.3 Same-product peer prices move during unchanged focal quotations

All twenty same-source unchanged runs longer than 36 observations coincide with at least one updating peer county. Three also coincide with amplitude above 20% in the median of the other eleven counties. Qingyuan urea from 25 March 2022 to 25 January 2024 has 67 focal observations and peer-median amplitude of 30.77%; Jianping urea from 15 April 2022 to 25 January 2024 has 65 observations and the same amplitude. Their complete-peer-date coverage is 67/67 and 65/65, and eleven and ten peers, respectively, update within the runs.

Dawa urea from 5 February 2024 to 15 March 2025 has 41 observations, a peer-median amplitude of 25.71%, ten updating peers and coverage of 40/41 dates. The peer median ranges from 1.30 to 1.70 CNY per 500 g in the Qingyuan and Jianping windows and from 1.05 to 1.32 in the Dawa window. The later 44-observation Jianping run has amplitude exactly equal to 20.00%, so it falls at the threshold. Quotations for the same product change elsewhere in Liaoning during every long unchanged focal quotation run.

7 Discussion

7.1 Benchmarks determine the historical question and the alerts

Benchmark selection and quotation history jointly determine the monitoring signal. Historical-window choice changes attainment and alert dates, while county screening changes the excess-incidence classification. The historical windows contain different relative-price conditions. The 2018–2019 comparison classifies almost every complete later date for all twelve counties as attained, whereas the 2020 comparison classifies roughly one-third. Benchmark choice therefore determines which dates enter county-incidence summaries and which trigger an alert.

The two references answer different questions. The fixed reference varies historical county deviations along the observed aggregate trajectory. The paired experiment also varies benchmark means and attainment dates. For the ten-county group, the 2019–2020 fixed-reference tail estimate is 0.010. Adding historical-mean variation produces a paired interval containing zero and changes the excess-incidence classification.

All twelve counties have strictly positive paired intervals under all five benchmarks. Removing Jianping and Qingyuan produces a positive interval for the one-year 2020 benchmark and intervals containing zero for the other four. Removing all six flagged counties produces intervals containing zero under every benchmark in the 1,999-draw comparison. The selected 9,999-draw checks for 2018–2020 and 2020 give the same classification. County membership therefore determines which benchmark comparisons show excess incidence.

7.2 Quotation persistence links historical levels to monitoring signals

The observation basis links a repeated quotation to its market meaning: a current price, an earlier quotation, or a sampling point awaiting new information. The agricultural-input rules explicitly distinguish a current transaction, an available-stock posted price and the most recent transaction for an unavailable item (Price Monitoring Center, 2015). The CPI Manual requires complete re-recording even when a price is unchanged and describes the effect of carrying old observations forward (ILO et al., 2020, paragraphs 5.65 and 6.61).

The county patterns identify different priorities for checking that record. Jianping has low numerical updating in both products, directing attention to the county's reporting and sampling-point history. Qingyuan's maize rate exceeds the maize low-update cutoff, while its urea rate is below the urea cutoff, directing attention to the product record. Jianping's unchanged 1.50-CNY urea value crosses the January–February 2024 source-system change with the same specification and unit. Its later gap separates the 94-observation segment from another 44 equal observations. National-turn matching records dispersed update dates and changes with the monthly coverage rule. The verification unit is therefore a county–product observation with its sampling enterprise, price basis and transaction date.

Persistent quotations also enter the benchmark itself. Four of the six representative runs overlap 2018–2020, and Jianping's three-year reference combines long periods at two urea levels. During a later unchanged-urea segment, all recorded exchange-ratio movement comes from maize, while the earlier urea quotations determine the historical comparison level. The reporting history therefore affects both threshold construction and later classifications. Liaoning's documented trial reporting, site visits and monitoring-point replacements provide an administrative route for checking the flagged histories (Liaoning Provincial Development and Reform Commission, 2022).

7.3 Thresholds change monitoring workload

A one-third share rule requires four of ten counties, or 40%; three of ten represents 30%. A common 50% threshold aligns the minimum triggering share across the three groups, and substantial benchmark differences remain. For the

six screened counties, the 2020 window generates aggregate alerts on 131 dates, and the 2018–2019 window generates zero. The historical window therefore determines the volume of alerts generated for review.

Fewer alert dates can accompany more episodes when a higher threshold splits a long trigger run. This occurs for the twelve-county sample under the 2019–2020 benchmark and the six-county sample under the month-specific benchmark. Reporting dates and episodes together shows both the duration and recurrence of a monitoring signal.

Changing county membership changes both the denominator and the aggregate path. Under the 2018–2020 benchmark, removing the two low-update counties reduces aggregate alerts from nineteen to twelve monitoring dates; under the 2020 benchmark, applying the full screen increases them from 126 to 131. Monitors should display the county group alongside each alert series.

7.4 Contribution to agricultural price information

The World Bank’s affordability indicator compares an input–output price relationship with a stated historical average (World Bank, 2024); our county results show how alternative historical averages change local incidence. Baquedano’s (2015) anomaly indicator assesses seasonal and annual price growth; our alert comparison holds observations fixed and varies the historical threshold and sample membership. Eichenbaum et al. (2014) examine apparent small changes associated with measurement error; our long unchanged sequences identify observation records for follow-up. Galtier et al. (2014) discuss the design and impacts of agricultural market information systems; our date and episode counts show a consequence of design choices for potential monitoring workload.

8 Measurement basis and screening interpretation

We measure the quoted exchange terms between maize and urea in the twelve-county archive. Fertilizer expense as a share of maize revenue equals $(u/Y)/Q$; a lower exchange ratio raises this share for a given fertilizer-to-output ratio u/Y . The screen identifies county–product records for checking against sampling-enterprise, transaction-date and observation-status information. The inventory covers 31 exact product/specification/unit groups, with capture coverage recorded for each group and matched national comparisons for maize purchase and domestic urea. We compare proportional movement within each series to accommodate differences in product specifications and survey scope. Under partial coverage, the observed national amplitude provides a lower bound on the amplitude over the full matched window.

We developed the quotation screen after exploratory inspection of the data and applied the same rules to all twelve counties. The 36-setting grid quantifies the effect of threshold choice: Zhangwu is flagged at 10% and 20% amplitudes and remains unresolved at 30%. The 2020 benchmark uses 36 monitoring dates from the year immediately preceding the fertilizer-price surge. Its positive ten-county paired interval identifies excess incidence relative to that year’s exchange terms. Within-year resampling diagnostics and extended draws document the numerical precision of this comparison.

9 Conclusions

The results show that historical benchmark choice determines the monitoring classifications and alert frequency produced from the same county quotations. Across five benchmarks, attainment for all twelve counties ranges from 34.31% to 99.02%, while aggregate alerts for the six screened counties range from zero to 131 dates. The differences persist under a common 50% county-share threshold and all twelve symmetric county deletions. Choosing the historical window sets both the comparison level and the workload generated by monitoring alerts.

Persistent quotations connect historical benchmark levels to later county signals. The combined screen flags six of twelve counties. Jianping records 109 identical urea quotations over 1,096 days, and Jianping and Qingyuan contribute 54.64% of accumulated shortfalls under the 2018–2020 benchmark. Four representative runs overlap that window, linking quotation histories to the benchmark itself. Counties flagged from historical observations continue to update less frequently in both later periods; every long unchanged run coincides with an updating same-product peer elsewhere in the province. These findings identify county–product records for checking the sampling point and the date and basis of the reported quotation.

County membership changes the classification of excess incidence. All five paired intervals are positive for the original twelve counties. Excluding Jianping and Qingyuan produces a positive interval under the one-year 2020 benchmark and intervals containing zero under the other four; the full screen produces intervals containing zero under every benchmark. These results identify quotation history and county composition as determinants of the distributional signal. Monitoring reports should show aggregate attainment beside county incidence and report alert dates together with episodes. Stating the reference window, county membership, update history and trigger threshold makes the resulting signals comparable and identifies the quotation records to check first.

Appendix A Supporting definitions and quotation changes

A.1 Calendar alignment and paired resampling

The fixed-reference paths cover all 207 planned later observations before the original 204-complete-date mask is applied. The historical residual vector includes every county in the comparison group. Annual benchmarks use non-circular blocks within their own historical windows. For the month-specific benchmark, each block matches both its starting month and its fifth/fifteenth/twenty-fifth reporting position, then follows the historical seasonal sequence. The seasonally aligned outer blocks fill the historical target calendar, and the inner residual paths use the same alignment on the later calendar. County groups share date indices within a benchmark and block length and recalculate their own aggregate path, attainment set and denominators.

In a paired draw, the resampled history first supplies new county means. Historical residual candidates are then recomputed and centered across counties on each donor date. One independent residual path is sampled for that draw, and both incidence statistics are evaluated on the same recalculated attainment set. We calculate incidence statistics for draws with at least one attainment date. The mean difference, its interval and the positive-draw fraction summarize the joint distribution of historical-mean variation and residual-path variation. The positive fraction reports the share of paired differences above zero; the fixed-reference tail estimate counts reference statistics at least as large as the observation.

A.2 Tail estimates and numerical precision

The fixed-reference estimate is $\hat{p} = (K + 1) / (R + 1)$, including ties. Its displayed resolution with 1,999 valid draws is 0.0005. At a zero tail count we display ≤ 0.0005 †; at a complete tail count we display ≥ 0.9995 †. The underlying plus-one values at those two boundaries are 0.0005 and 1.0000. A one-sided 95% exact binomial bound is about 0.00150 for a zero tail and about 0.99850 for a complete tail. These bounds describe numerical tail-probability precision.

For interior counts, the Monte Carlo standard error uses the sampled tail fraction K / R . Percentile intervals use linear interpolation. The fixed-reference interval is two-sided 90%, and the paired interval is two-sided 95%. A rounded zero endpoint is evaluated using the full-precision result; exact zero belongs to the equality boundary. The selected 9,999-draw comparisons are reported separately from the uniformly 1,999-draw main table.

A.3 All six Jianping urea quotation changes

Table A1 records every source-continuous numerical urea update in Jianping. National extremum months and confirmation-publication dates are separate fields. We match each local update to national turns confirmed by its monitoring date.

Table A1. Jianping urea updates and the primary national-turn comparison.

Local monitoring date	Quote change (CNY/500 g)	National extremum month and confirmation published	Months from extremum	Within 3 / 6 months
2018-04-25	0.75 → 1.00	No eligible preceding turn	—	No / No
2021-05-05	1.00 → 1.10	2020-09; 2020-12-04	8	No / No
2021-08-15	1.10 → 1.40	2021-03; 2021-06-04	5	No / Yes
2022-04-15	1.40 → 1.50	2021-12; 2022-03-04	4	No / Yes
2026-02-25	1.50 → 1.20	2025-05; 2025-08-04	9	No / No
2026-09-05	1.20 → 1.10	2026-04; 2026-09-04	5	No / Yes

Note: The primary rule uses at least two observed dekads per month and a 5% cumulative change after two consecutive reversed monthly movements. Monthly gaps interrupt the sequence. Local dates are the archived monitoring dates; transaction dates and actual local publication dates are not recorded.

Five of six changes have an eligible preceding turn under all four matching rules. With the primary coverage rule, three occur within six months at the 5% threshold and two at 10%. Requiring all three dekads per month gives one six-month match at 5% and zero at 10%. All four settings produce zero three-month matches among the six updates. The same rules are applied to all 499 valid county urea changes, producing 1,996 matching records. These comparisons describe the timing of archived quotations against nationally published price information.

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Data availability: The complete data and code replication package accompanies this study and contains the normalized observations, archived source responses, saved reference results and analysis programs, including the county-influence, historical-screening and same-product peer results in Supplement S11. The project repository is <https://github.com/ByronLi0207/liaoning-price-monitoring>. The archive series is identified by concept DOI 10.5281/zenodo.23131960 (<https://doi.org/10.5281/zenodo.23131960>); the version-specific Zenodo record lists the deposited files and checksum.

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